

CSE110 Spring 2024 Lab Exam - 03 Tentative Solutions and Rubrics

In case of any circumstance not present in the rubrics, feel free to discuss in assigned slack channel. This rubric is provided to maintain a similar marking scheme throughout all sections.

[SET-A]

SOLUTION

```
import java.util.Scanner;
public class PasswordStrengthValidator {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input
        System.out.print("Enter password: ");
        String password = sc.nextLine();

        // Condition 1: Length check
        if (password.length() < 8) {
            System.out.println("Weak Password");
            return;
        }

        boolean hasUpper = false;
        boolean hasLower = false;
        boolean hasDigit = false;
        boolean hasSpecial = false;

        // Traverse each character of the password
        for (int i = 0; i < password.length(); i++) {
            char ch = password.charAt(i);

            if (ch >= 'A' && ch <= 'Z') {
                hasUpper = true;
            } else if (ch >= 'a' && ch <= 'z') {
                hasLower = true;
            } else if (ch >= '0' && ch <= '9') {
                hasDigit = true;
            } else {
                hasSpecial = true;
            }
        }

        // Final decision
        if (hasUpper && hasLower && hasDigit && hasSpecial) {
            System.out.println("Strong Password");
        } else {
            System.out.println("Weak Password");
        }
    }
}
```

RUBRIC

SL	Points To Meet	Marks [10]
1	Properly taking input	1
2	Declaring boolean variables	2
3	Traversing and checking in each segment	4
4	Combining final decision	2
5	Properly printing once	1

[SET-B]

SOLUTION

```
import java.util.Scanner;
public class EmailValidation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter email: ");
        String email = sc.nextLine();
        boolean isValid = true;
        int atCount = 0;
        int atPos = -1;
        // Check for spaces and count '@'
        for (int i = 0; i < email.length(); i++) {
            char ch = email.charAt(i);
            if (ch == ' ') {
                isValid = false;
            }
            if (ch == '@') {
                atCount++;
                atPos = i;
            }
        }
        // Must have exactly one '@'
        if (atCount != 1) {
            isValid = false;
        }
        // '@' cannot be first or last
        if (atPos == 0 || atPos == email.length() - 1) {
            isValid = false;
        }
        // Check for '.' after '@'
        boolean dotFound = false;
        for (int i = atPos + 1; i < email.length(); i++) {
            if (email.charAt(i) == '.') {
                dotFound = true;
                break;
            }
        }
        if (!dotFound) {
            isValid = false;
        }
        // Final output
        if (isValid) {
            System.out.println("Valid Email");
        } else {
            System.out.println("Invalid Email");
        }
    }
}
```

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RUBRIC

SL	Points To Meet	Marks [10]
1	Properly taking input	1
2	Declaring boolean and @ sign position variable	1
3	Checking for space and @ sign position	3
4	Checking for availability of dot (.) after @ sign	3
5	Properly printing final output	2